

自动驾驶感知算法简介

大纲

- 自动驾驶及感知算法发展历程
- 感知算法发展趋势
- 量产感知算法开发流程

自动驾驶发展历程

DARPA
AVL





CMU
ALVINN



帕尔马大学
ARGO项目



DARPA
Grand
Challenge

1984 → 1986 → 1998 → 2004

2005



DARPA
Grand Challenge

2009



Google自动驾驶成立

2013

自动驾驶迅猛发展







2015

小鹏汽车成立（2014）

2016

特斯拉推出AutoPilot



2017

Google自动驾驶改名Waymo，
与克莱斯勒合作量产

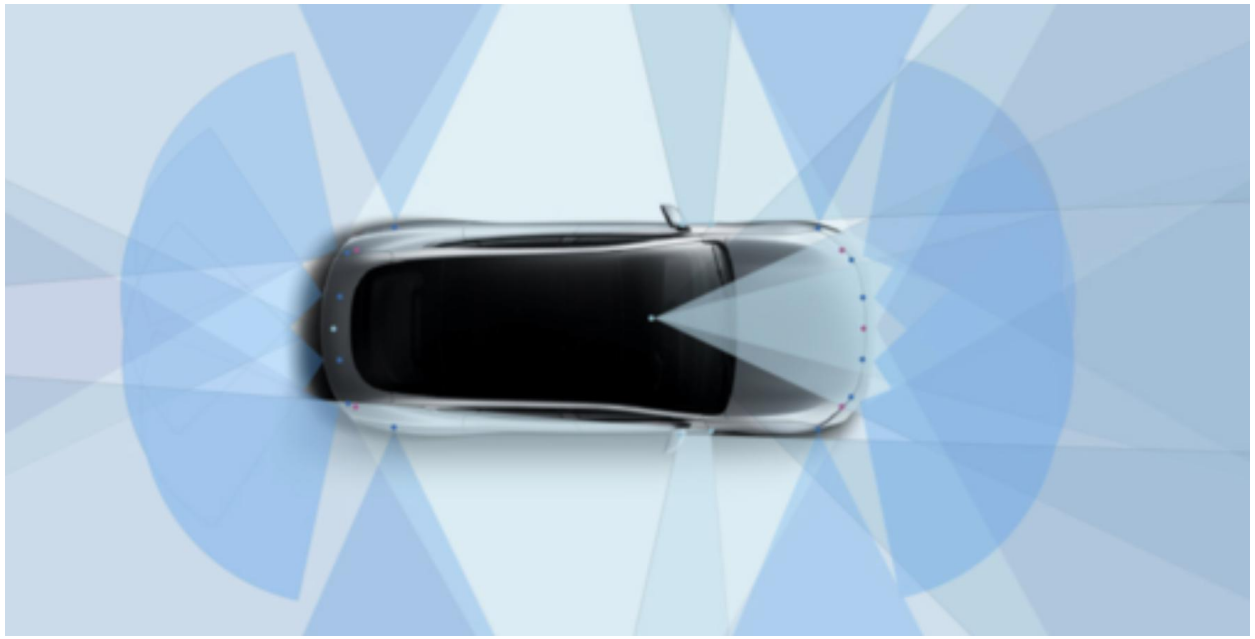


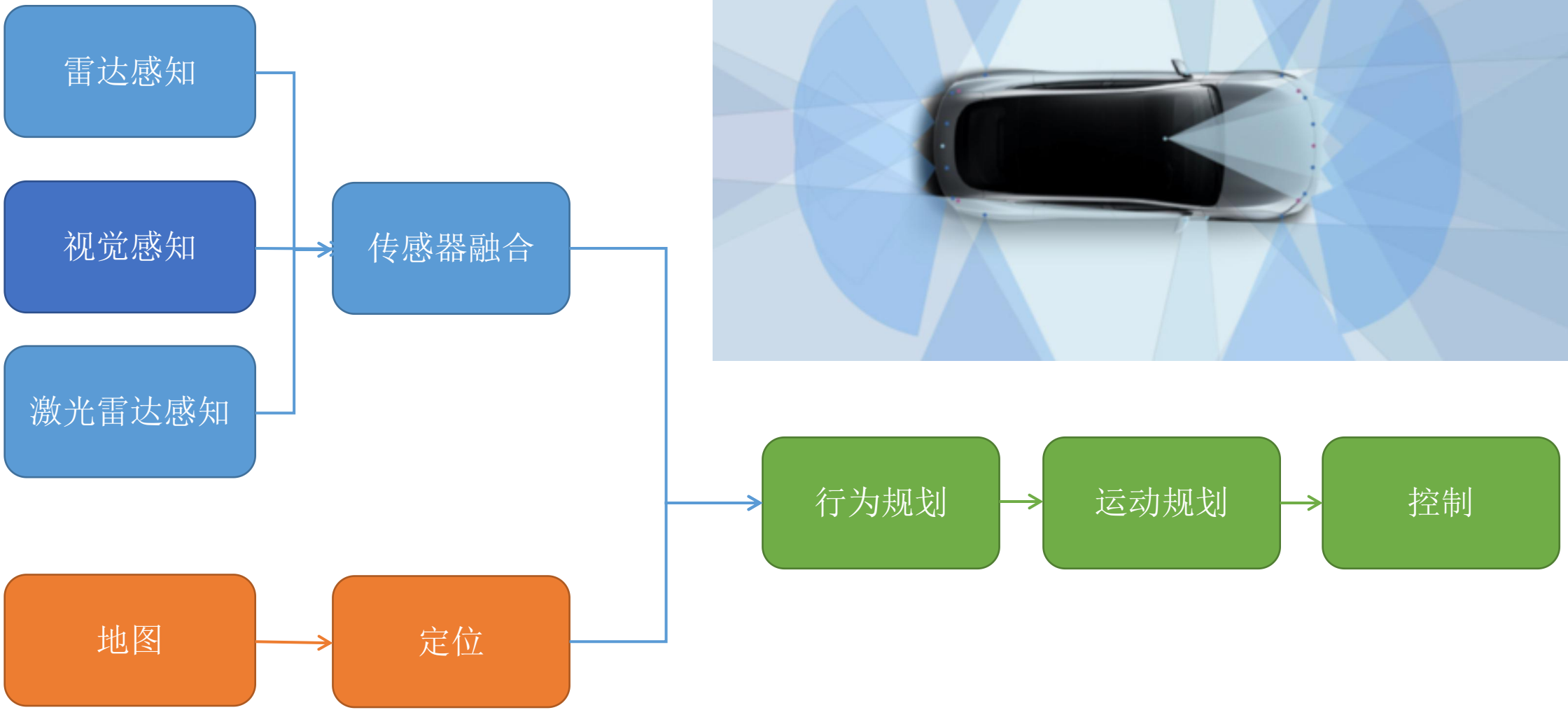
2020

小鹏P7上市



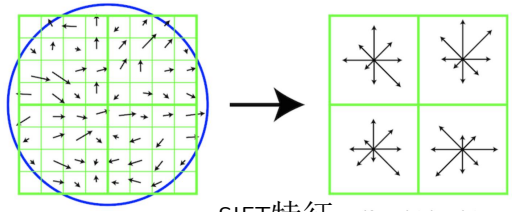
自动驾驶主流系统架构（简化）



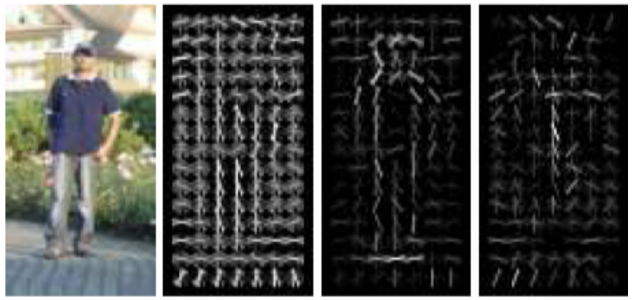


视觉感知算法发展历程

1990-2010
传统视觉方法: 特征+统计



SIFT特征

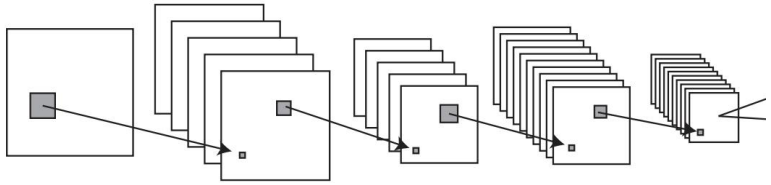


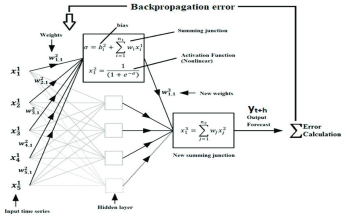
HOG特征

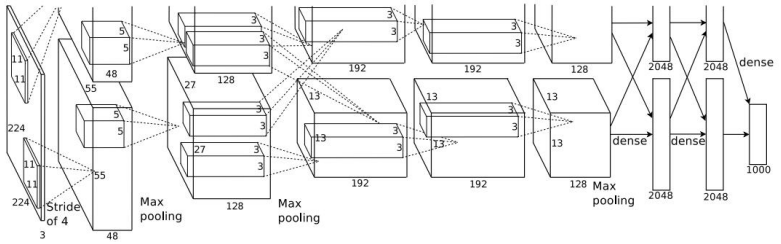


DAISY 特征

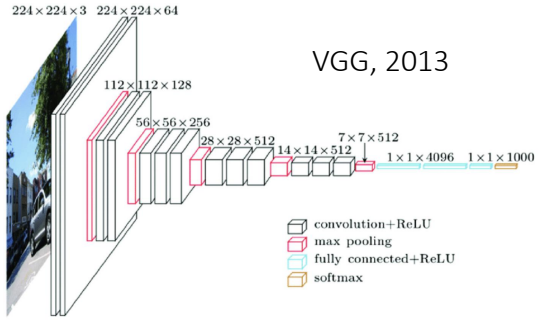
2010 - 目前
深度学习：卷积网络+反向传播



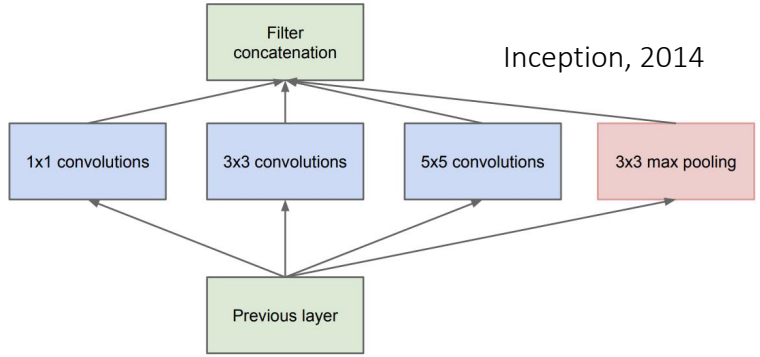




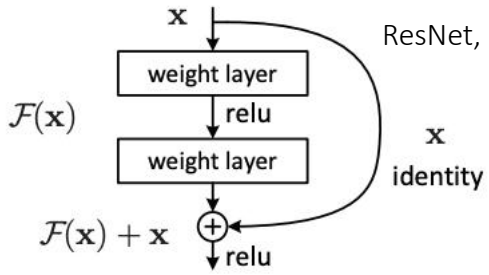
Alexnet, 2012



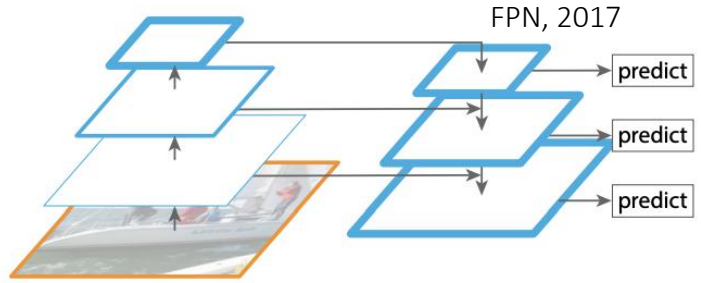
VGG, 2013



Inception, 2014



ResNet, 2015

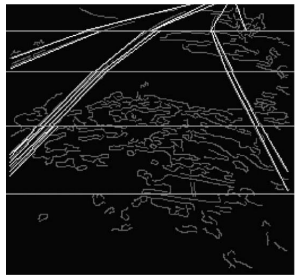


FPN, 2017

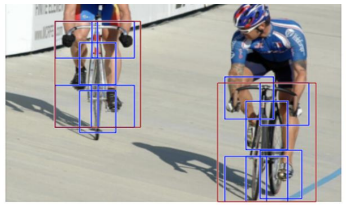
自动驾驶视觉感知算法发展趋势

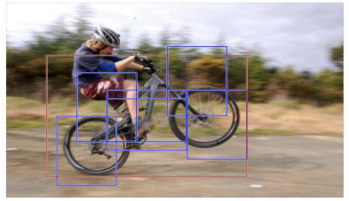
1990-2010
传统视觉方法

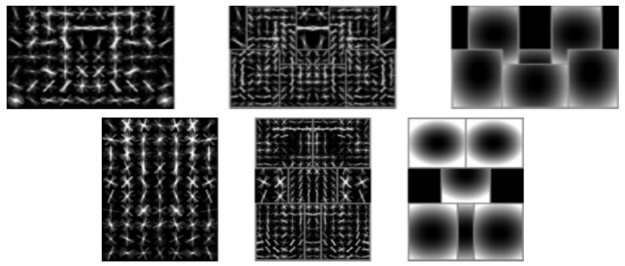




车道线B-Spline拟合

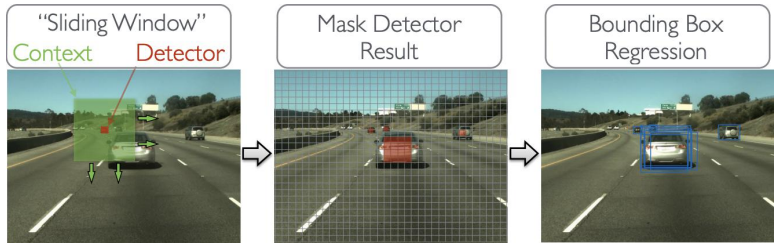


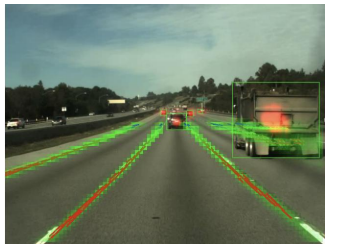




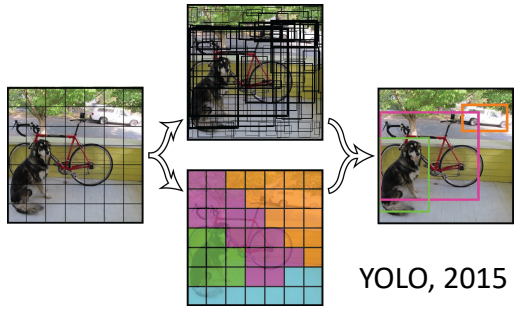
DPM 物体检测

2014 - 目前
深度学习

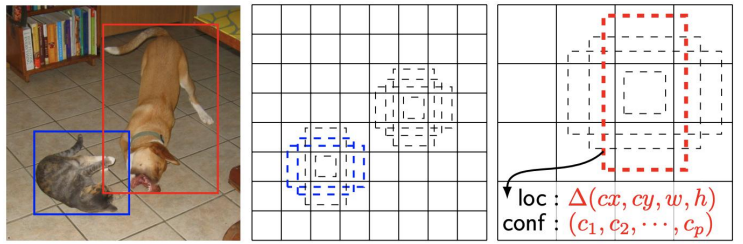




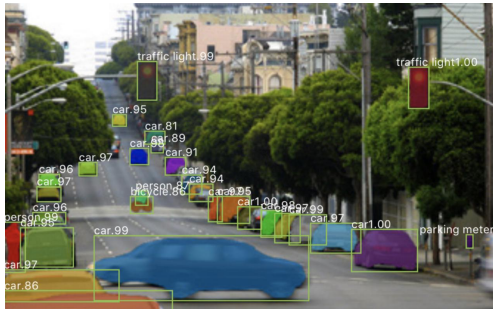
Stanford, 2015



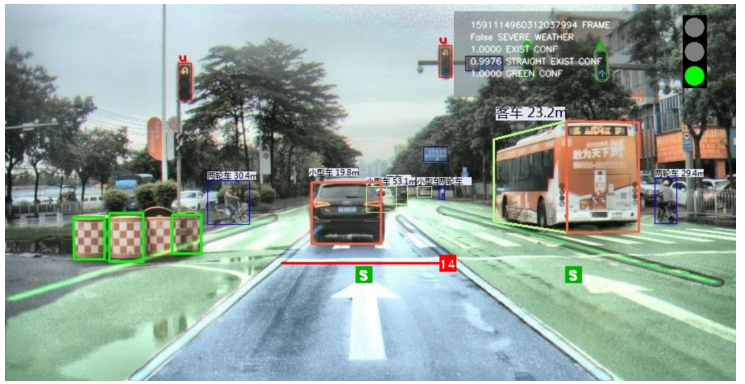
YOLO, 2015



SSD, 2016



MaskRCNN, 2017



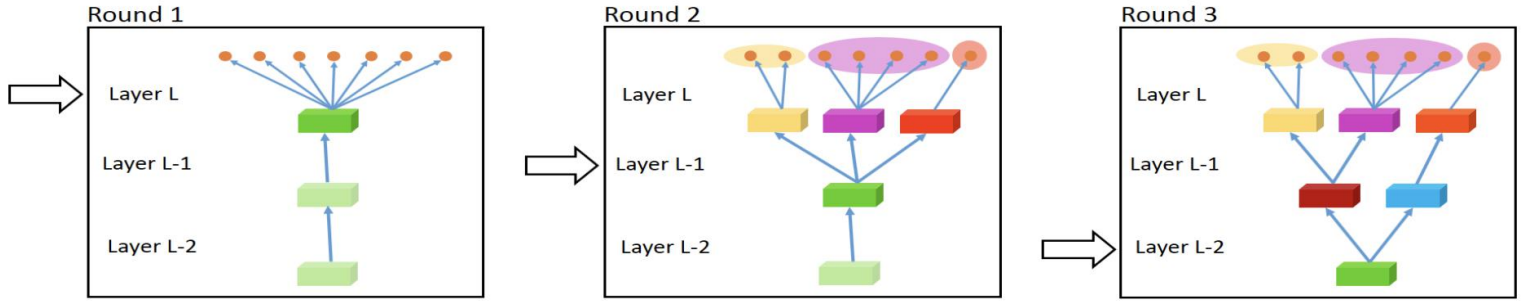
Xpilot， 2020

自动驾驶视觉感知算法发展趋势

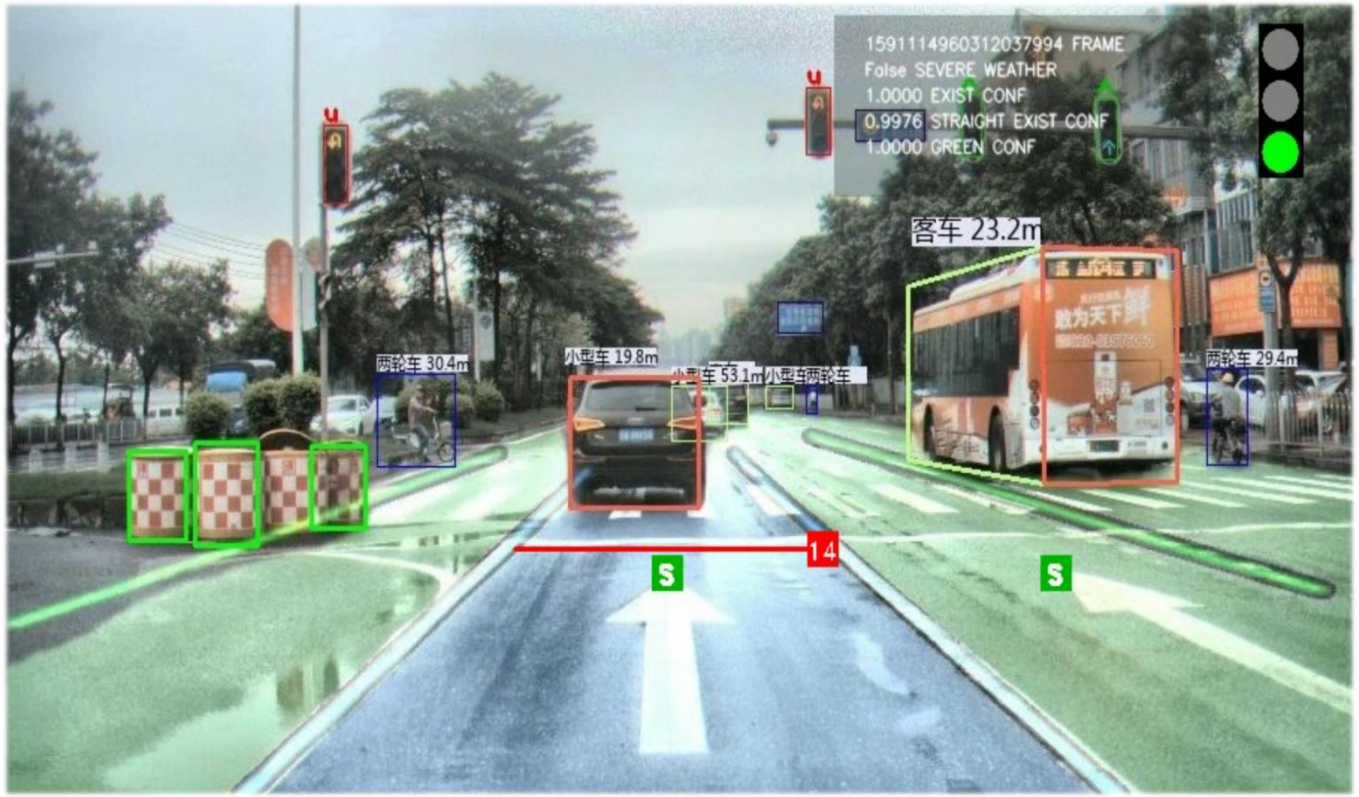
1. 任务多样化

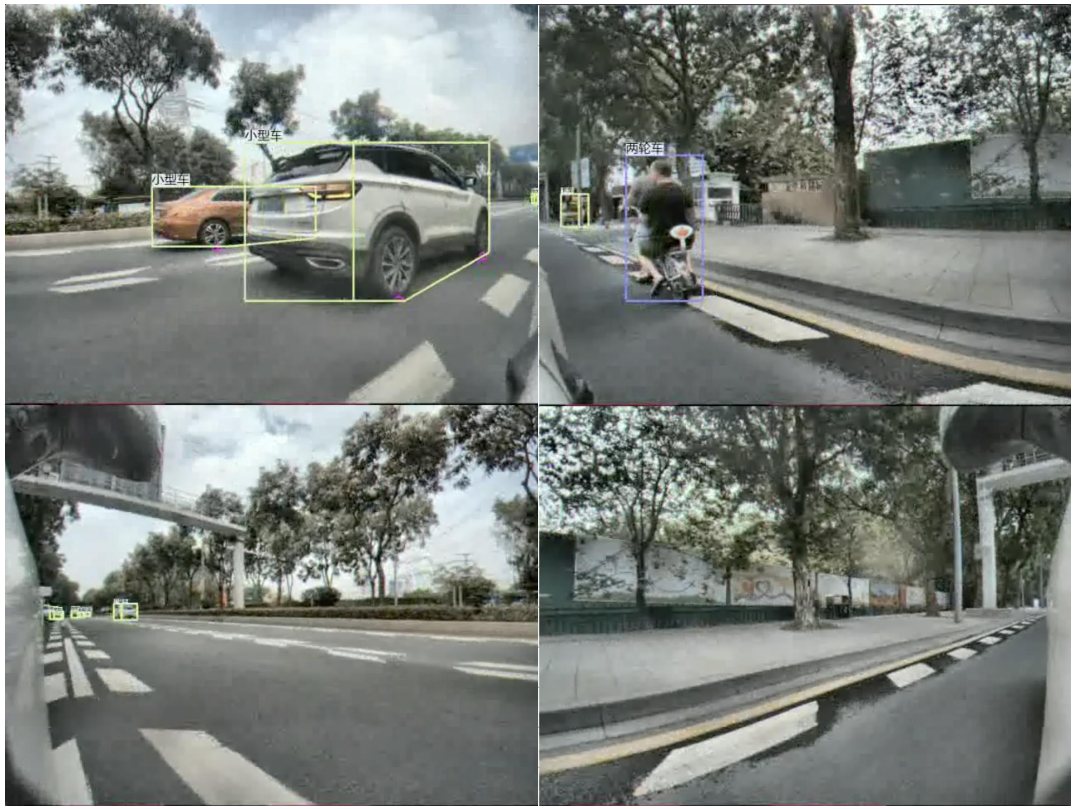
2. 输入时序化

3. 输出三维化



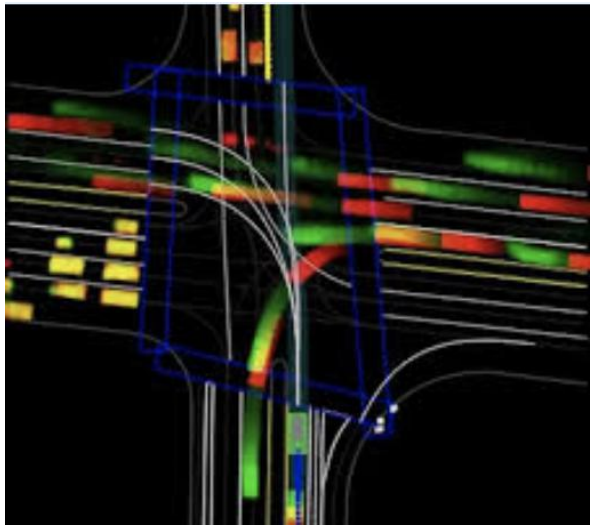
Fully-adaptive Feature Sharing in Multi-Task Networks, Lu, et al., 2016





自动驾驶视觉感知算法发展趋势

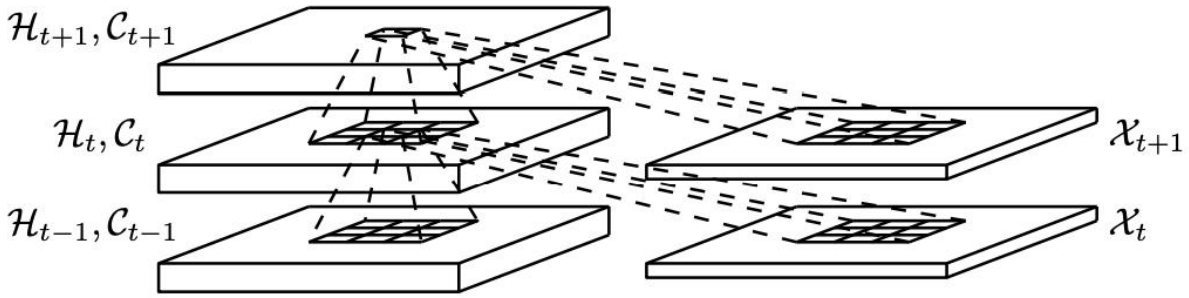
1. 任务多样化
2. **输入时序化**
3. 输出三维化



Chauffeurnet, 2018



Nvidia DriveWorks, 2020

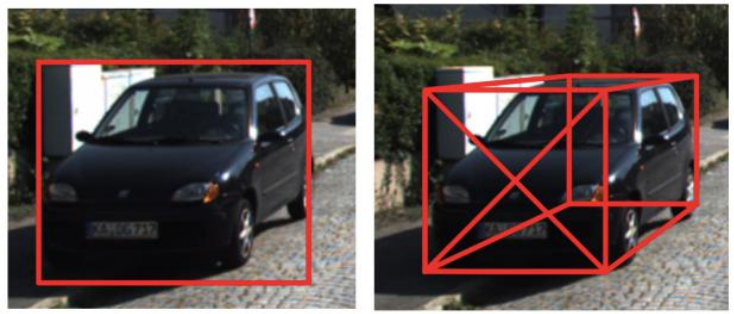


ConvLSTM, 2015

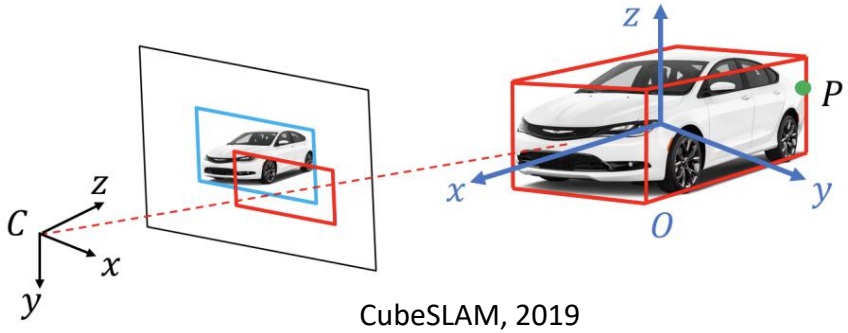


自动驾驶视觉感知算法发展趋势

1. 任务多样化
2. 输入时序化
3. **输出三维化**



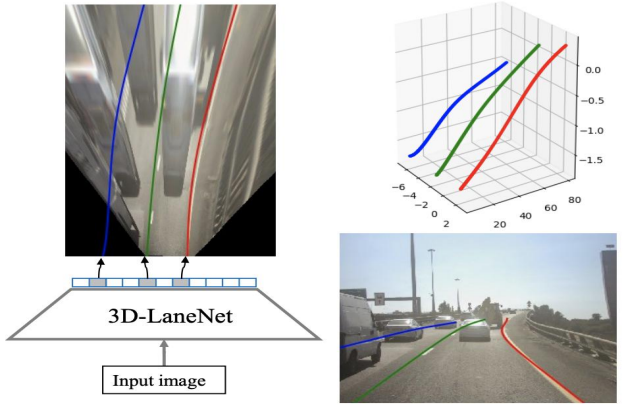
Deep3DBox, 2017



CubeSLAM, 2019

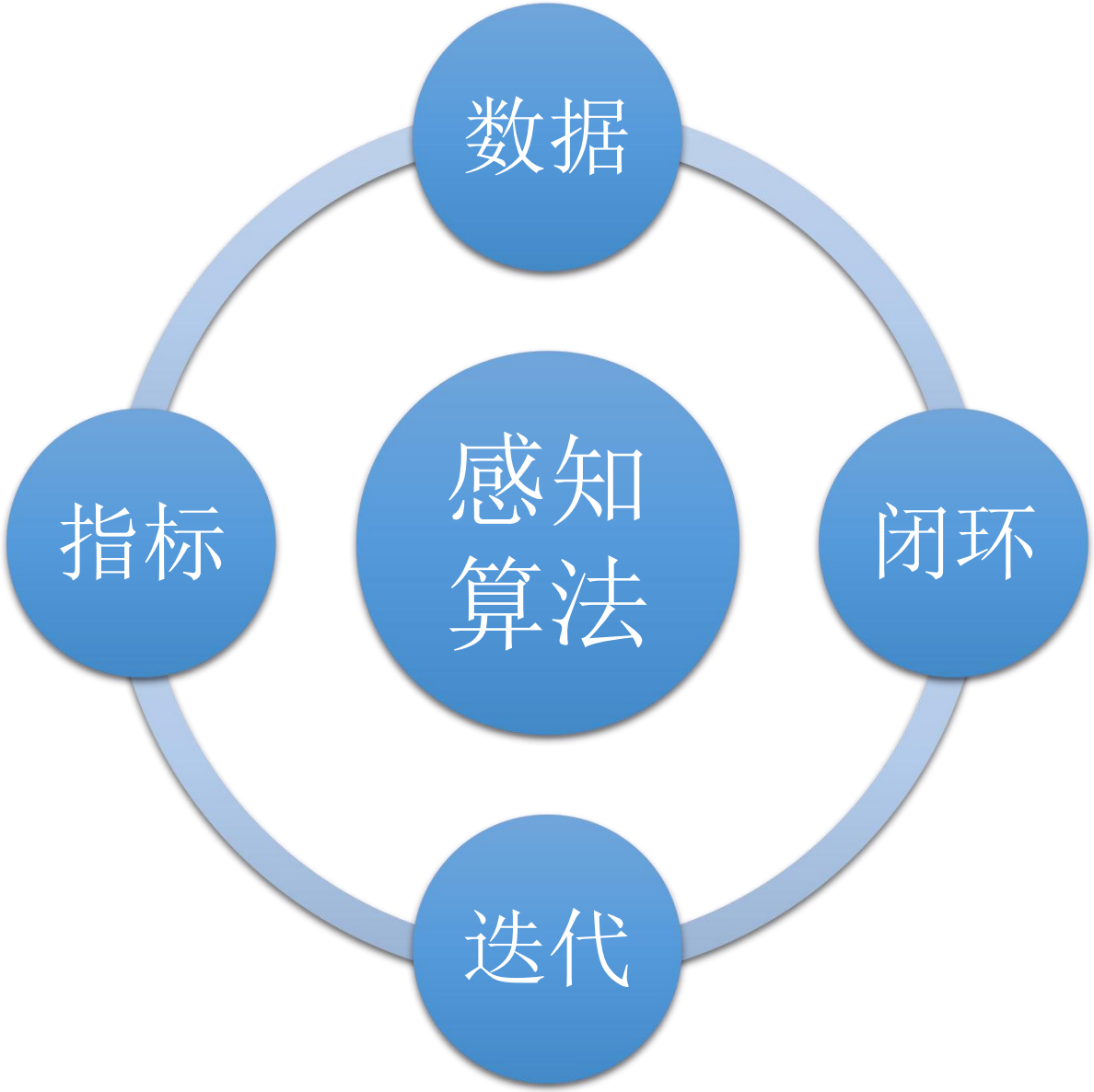


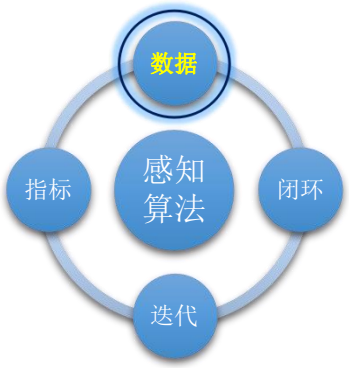
MonoLayout, 2020

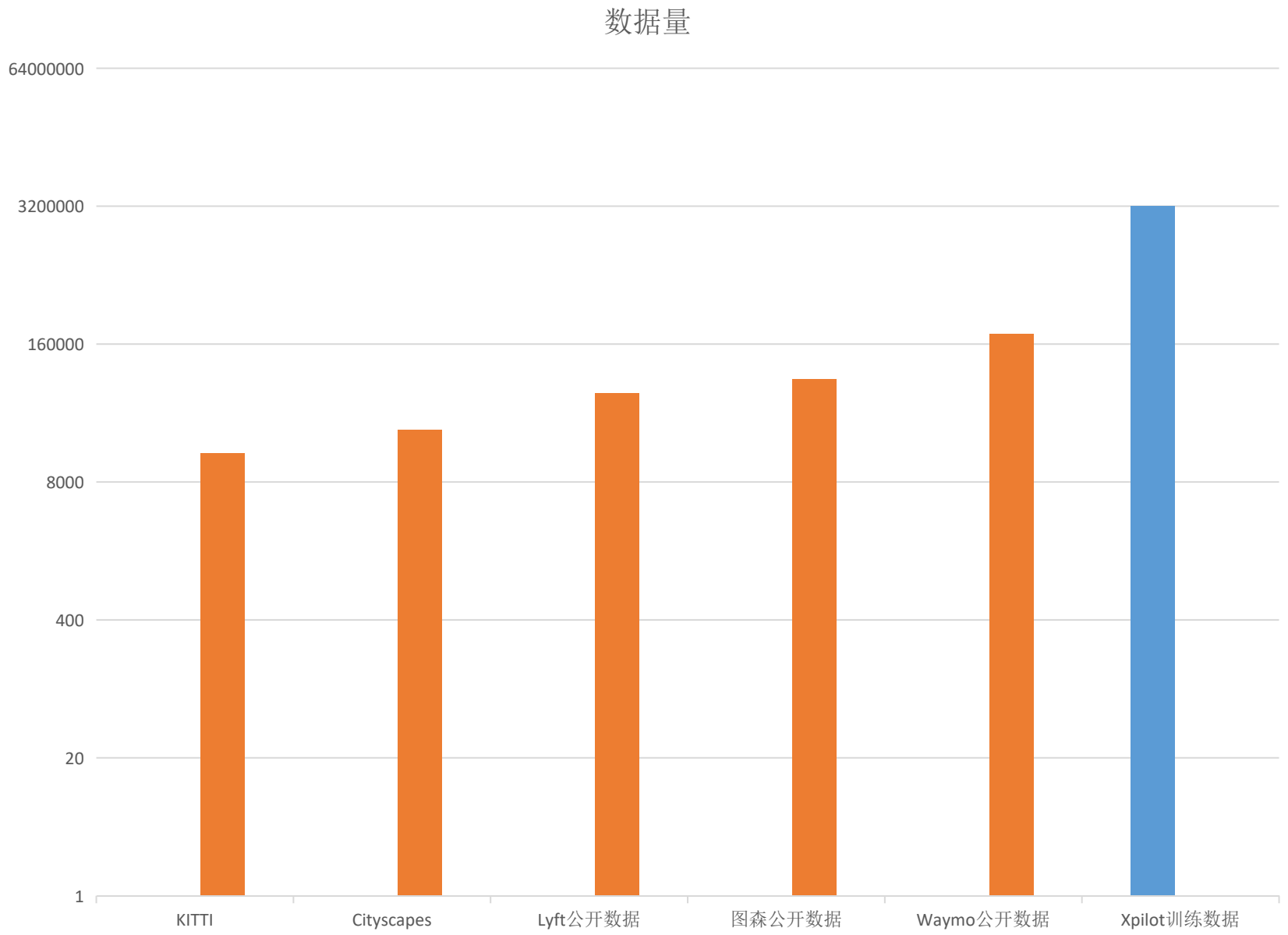


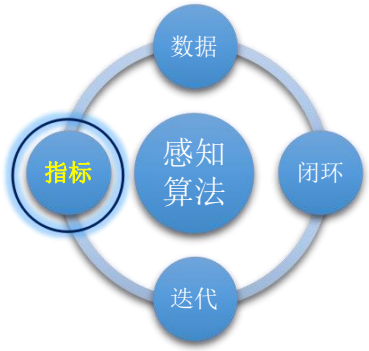
3D-LaneNet, 2019

量产感知算法开发流程




数据
指标
感知
算法
闭环
迭代


数据量
64000000
3200000
160000
8000
400
20
1
KITTI
Cityscapes
Lyft公开数据
图森公开数据
Waymo公开数据
Xpilot训练数据



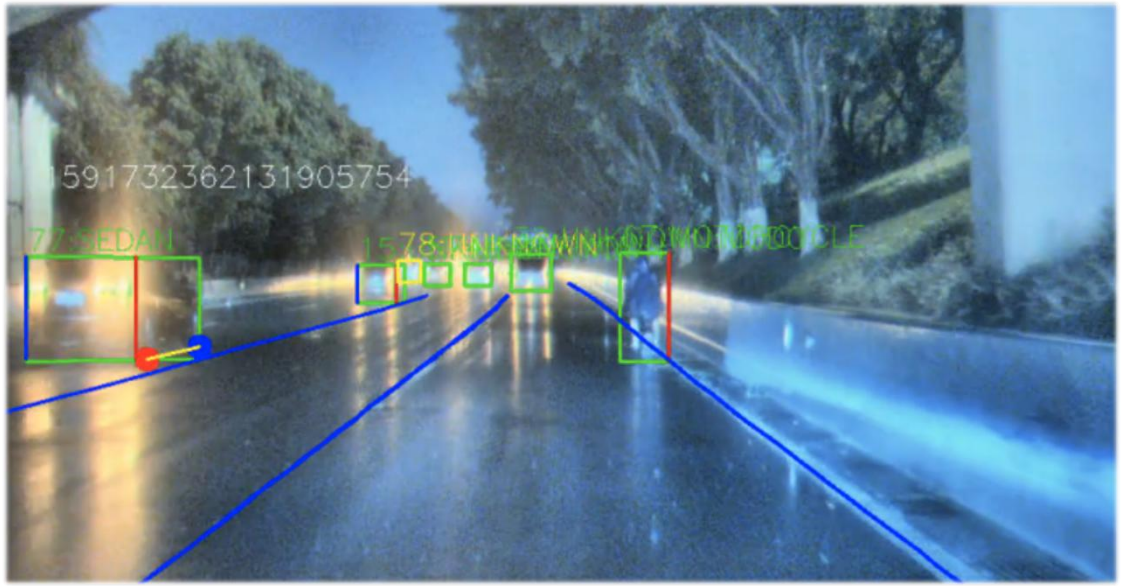
感知KPI
- 均方根误差 RMSE
- 2D IoU
- 3D IoU
- 航向角误差
- 时序一致性

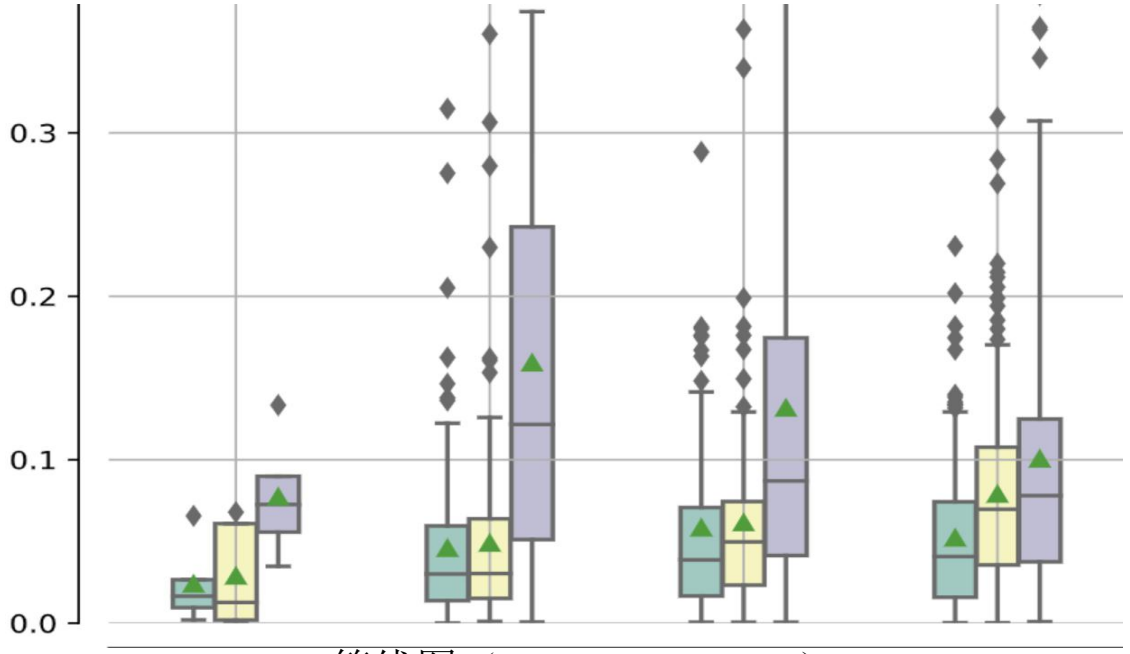
……

端到端KPI
- 百公里误检率
- 漏检率

……

- 选择更有物理意义的KPI
- 平均数不说明任何问题
- 关注百分位，关注最坏情况





箱线图（Box-Whisker Plot）





数据分类，清洗，标注



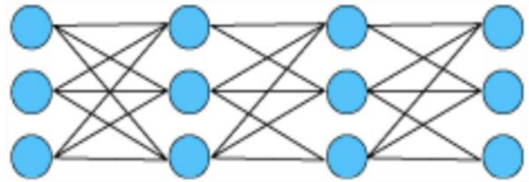


车载平台

数据上传

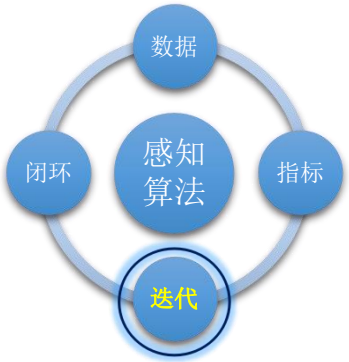


大数据平台



感知算法开发、验证

OTA 远程升级



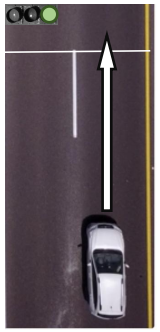
功能复杂度



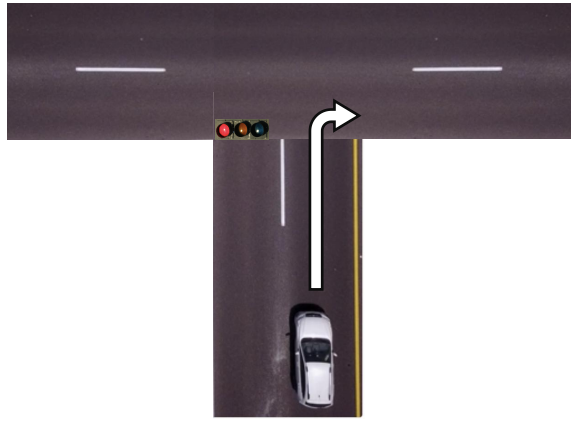
仪表盘显示



红绿灯语音提醒



主要路口自动直行



多数路口自动通过

95%

99%

99.99%

99.999%

算法性能


THANKS FOR YOUR ATTENTION
「通过智能智造，创造更美好的出行生活」